



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

TECHNICAL DRAWING INTERPRETATION

LSVDI302

Read and interpret Technical drawing

Competence

RQF Level:

3

Learning Hours



70

Credits:

7

Sector:

Construction and Building Services

Trade:

Land Surveying

Module Type:

Specific

Curriculum:

CBSLSV3001- TVET Certificate 3 in Land Surveying

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Purpose statement	This module describes the skills, knowledge, and attitude required to interpret technical drawings. By the end, the student will be able to draw different types of lines and plain geometry, apply projection in 2D and 3D Drawings, to interpret both architectural drawings and road layout drawings that could be used by the surveyor to perform his/her job.					
Leaning assumed to be in place	Safety, Health and environment at workplace; Surveying tools and equipment; Surveying communication language.					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	• Group project and presentation	20%				
	• Individual project /Work	50%				
			Summative Assessment			50%

Elements of Competency and Performance Criteria

Elements of competency	Performance criteria
1. Select materials, tools, and equipment	1.1. Materials are properly identified in accordance with the construction drawings.
	1.2. Tools and equipment are properly distinguished in accordance with the construction drawings.
	1.3. Materials, tools, and equipment are properly well maintained according to their types.
2. Draw lines, symbols, Plain geometry and letters	2.1. Lines used in construction drawings are correctly identified according to their types.
	2.2. Construction drawing symbols are properly identified according to their types.
	2.3. Lines, figures, and curves are accurately drawn according to their types.
	2.4. Lettering techniques are well applied in accordance with ISO3098-1:2015.
3. Interpret drawing scale and	3.1. Types of scales are well identified as use in construction

dimensions	drawing.
	3.2. Drawing scales are accurately used according to their types.
	3.3. Types and methods of dimensioning are clearly identified as use in construction drawing.
4. Perform projection	4.1. Methods of projection are correctly identified as use in construction drawings.
	4.2. Types of projection are properly identified as used in construction drawings.
	4.3. Views are accurately produced according to the types of projection.
5. Interpret building plans, views and section	5.1. Site plan is properly interpreted according to the drawing specifications.
	5.2. Floor and foundation plans are properly interpreted according to the drawing specifications.
	5.3. Building views and sections are well interpreted according to the drawing specifications.
	5.4. Electrical, plumbing, and drainage plans are properly interpreted according to the drawing specifications.
6. Interpret road layout drawings	6.1. Elements of road cross-sectional are properly identified according to the type of road.
	6.2. Road layout drawing is properly interpreted according to the drawing specifications.
	6.3. Road profiles are accurately interpreted according to the drawing specifications.
7. Present drawing sheet	7.1. Elements of the drawing sheet are properly identified in accordance with the construction drawings.
	7.2. Sizes of drawing sheet frame and margin are well-identified according to ISO.
	7.3. Elements of the title block are clearly identified according to the type of project.

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Select materials, tools, and equipment 2. Draw lines symbols and plain geometry 3. Interpret drawing scales and dimensions 4. Perform projection 5. Interpret building plans, views, and section 6. Interpret road layout drawings 7. Present of the drawing sheet
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Learning outcome 1: Select materials, tools and equipment		Learning hours: 4
Indicative content		
* Identification of drawing material (Refer to ISO 2692) • Ink, lead mine, paper, eraser * Identification of drawing tools and equipment • Tracing tools, writing tools, accessories * Maintenance of drawing materials, tools, and equipment • Cleaning of drawing tools and equipment • Storage of drawing tools • Care and safety of drawing tools		
Resources required for the learning outcome		
Equipment	• Drawing table, projector, and laptop.	
Materials	• Ink, lead mine, pieces of paper, eraser.	
Tools	• Ruler, set squares, protractor, T-square, compass, divider, triangular scale, pencil, lead mine pens, stencils, French curve, template, pencil sharpener, drawing pin, and clips.	
Facilitation techniques	• Individual and group work • Group discussion • Brainstorming	
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment	

Learning outcome 2: Draw lines symbols and Plain geometry	Learning hours: 18
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Indicative content	
* Identification of lines used in construction drawing • Types of lines • Sizes of lines • Application of lines styles/types in construction drawing * Identification of types of symbols used in construction drawing (Refer to ISO 1101) • Basic masonry structural symbols • Basic plumbing and pipeline layout symbols • Basic drainage layout symbols • Basic electrical layout symbols • Basic road layout symbols * Drawing of lines, figures, and curves • Polygons, tangent, curves, cycloid, involute, spiral, ellipse, parabola, hyperbola * Application of lettering techniques • Lettering proposition • Order of strokes • Sizes of letters • Lettering procedure	
Resources required for the indicative content	
Equipment	Drawing table, projector, laptop
Materials	Papers, lead, eraser
Tools	Ruler, set squares, protractor, T-square, compass, divider, triangular scale, pencil, lead mine pens, pencil sharpener, drawing pin and clips.
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

Learning outcome 3: Interpret drawing scales and dimensions		Learning hours: 4
Indicative content		
* Identification of type of scales • Metric scale • Representative fraction scale • Graphical scale * Proper use of drawing scale • Full size, reduced size, enlarged size * Identification of types and methods of dimensioning • Principals of dimensioning • Types of dimensioning (aligned, linear, angular, reference, and radius/diameter) • Methods of dimensioning (parallel dimensioning, superimposed running dimensioning, chain dimensioning, combined dimensioning)		
Resources required for the indicative content		
Equipment	Drawing table, projector, laptop.	
Materials	Pieces of papers, lead, eraser.	
Tools	Ruler, set squares, protractor, T-square, compass, divider, triangular scale, pencil, lead mine pens, pencil sharpener, drawing pin and clips.	
Facilitation techniques	• Demonstration and simulation • Practical exercise • Group discussion	
Formative assessment methods	• Written assessment • Performance assessment	

Learning outcome 4: Perform projection		Learning hours: 16
Indicative content		
* Identification of methods of projection • Line of sight vs plane of projection • First angle projection • Third angle projection * Identification of types of projection • Parallel projection (oblique and orthographic projections: axonometric and multi-view projection)		

- Perspective projection (linear and aerial perspectives) * Production of views - Front view, back views, right view, left view, top and bottom views	
Resources required for the indicative content	
Equipment	Drawing table, Projector, laptop.
Materials	Pieces of paper, lead, eraser.
Tools	Ruler, set squares, protractor, T-square, compass, divider, triangular scale, pencil, lead mine pens, pencil sharpener, drawing pin and clips.
Facilitation techniques	- Demonstration and simulation - Individual and group work - Practical exercise - Group discussion
Formative assessment methods	- Written assessment - Performance assessment - Product based assessment

Learning outcome 5: Interpret building plans, view and section	Learning hours: 14
Indicative content	
* Interpretation of site plan - Dimensions, setbacks, scale, labels, notes, legend, orientation, references features * Interpretation of floor plan and foundation plan - Walls, columns, footings, and beams dimensions, scale, and details * Interpretation of views and sections - View: ✓ Front side, backside, right side and left side - Section: ✓ Source of section ✓ Levels of building structural elements ✓ Height of building structural elements ✓ Internal construction of building elements * Interpretation of electrical, plumbing, and drainage plans ✓ Electrical Plan: location of manholes, electrical poles, underground cable line, sport light	

✓ Plumbing plan: location of manholes, pipelines, valves, water tanks, water tap ✓ Drainage plan: location of water channels, manholes, pipelines, septic tanks, soak way pit	
Resources required for the indicative content	
Equipment	Drawing table, projector, laptop.
Materials	Drafting tape/scotch, flip chart.
Tools	Floor plans, electrical plans, plumbing plans, pine, clip, drawing pen, ruler, markers.
Facilitation techniques	• Individual and group work • Practical exercise • Individualized • Group discussion
Formative assessment methods	• Written assessment • Performance assessment

Learning outcome 6: Interpret road layout drawings	Learning hours: 9
Indicative content	
* Identification of elements of road cross-section • Right of way, formation with, carriageway, shoulder, road margin, camber, curbs longitudinal drain, building line, control line, culvert, dish channel * Interpretation of road layout drawings • Layout of Centreline, carriageway width, shoulder width, total road width, drained rain, road setbacks, tangent, and curves. * Interpretation of road profiles • Longitudinal profile: Ground elevation, datum elevations, project elevation, height difference, partial distance, cumulative distance, alignment, straight grade lines, and curve.	

• Transverse profile/ cross-sections: cross slope, road lanes, sidewalks, drainage features, pavement structure	
Resources required for the indicative content	
Equipment	Drawing table, projector, laptop
Materials	Drafting tape/scotch, flip chart
Tools	Road layout plans, profiles drawings, cross-sectional drawing, pine, clip, drawing pen, ruler, markers
Facilitation techniques	• Demonstration and simulation • Individual and group work • Practical exercise • Individualized • Trainer guided • Group discussion
Formative assessment methods	• Written assessment • Oral assessment • Performance assessment • Product based assessment

Learning outcome 7: Present drawing sheet	Learning hours: 5
Indicative content	
* Identification of elements of drawing sheet layout • Drawing frame, drawing margin, title block, and drawing area * Identification of sizes of drawing sheet frame and margin • Size of drawing sheet frame: A0, A1, A2, A3, A4 • Size of Margin of drawing sheet: Filled margin and unfilled margin * Present the main elements of the title block • Name and address of the company or design activity • Title of the project • Drawing name • Drawing number, sheet number • Location of the project • Names and dates of the drafters, checker, and issue date	

- Drawing scale

Resources required for the indicative content

Equipment	Drawing table, projector, laptop.
Materials	Pieces of paper, lead, eraser.
Tools	Ruler, set squares, protractor, T-square, compass, divider, triangular scale, pencil, lead mine pens, pencil sharpener, drawing pin and clips.
Facilitation techniques	• Individual and group work • Practical exercise
Formative assessment methods	• Written assessment • Performance assessment • Product based assessment

Integrated/Summative assessment (For specific module)

INTEGRATED SITUATION

HORIZON CONSTRUCTION Company is hiring a well-qualified company that will consult a contracting company on the construction of the pharmaceutical industry in Rwanda Special Economic Zone located in Masoro, Gasabo district, city of Kigali. As a land surveying technician of the consulting company, you are requested to read and interpret the drawings given and provide the following information and documents:

1. Number of columns presented in the floor plan
2. Number of spotlights presented on-the site plan
3. The levels of beams and roof
4. Sizes (width, length and height) of given footing (example footing A3)
5. Overall dimension of the backside of the building
6. Building Set back to the center of the road
7. Distance between Manhole A and Manhole B
8. Class of surrounding features
9. Elements of road drawn on the given plans
10. Draw a curve joining the Center-lines of the two adjacent roads shown on the drawing sheets and dimension the drawn curve by using the appropriate method

11. Draw a front view of the building based on the perspective drawings given

Note: Front elevation should be drawn on the separate sheet containing all required information for the drawing sheet presentation

Resources

Tools	Drawing sheet, drawing pens, protractor, ruler, set square, compass, divider.
Equipment	A well-equipped drawing-room.
Materials/ Consumables	Pieces of paper, eraser, leads.

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome 1: (10%)	1.1. Materials are well selected.	1. Materials are well selected.			1
		2. Materials are well used.			1
	1.2. Tools and Equipment are well Selected.	1. Tools are well selected.			2
		2. Tools are well used.			2
		3. Equipment is well selected.			1
		4. Tools are well used			2
		5. Equipment is well used.			1
	1.3. Tools and equipment are well maintained.	1. Neatness of drawing sheet is observed.			2
		2. Tools are well cared.			2
		3. Equipment is well cared for.			1
Learning outcome	2.1. Lines are well interpreted.	1. Outlines are accurately identified.			2

2: (20%)		2. Centre line is accurately identified.			2
	2.2. Symbols used are well interpreted.	1. Masonry symbols are accurately interpreted`			5
		2. Electrical symbols are properly interpreted.			4
		3. Plumbing symbols are properly interpreted.			3
		4. Curve is accurately drawn.			9
Learning outcome 3: (5%)	2.1. Dimensions and scale are well interpreted.	1. Dimensions of the back site are accurately identified.			2
		2. Curves are well dimensioned.			3
		3. Drawing scale is accurately used.			
Learning outcome 4: (25%)	4.1. Front view is well drawn.	1. Building basement is well projected.			2
		2. Building corners are well projected.			7
		3. Openings are well projected.			7
		5. Roof is well projected.			6
		6. Sizes of building elements are well respected.			3
Learning outcome 5: (25%)	5.1. Floor plan and foundation plan are well interpreted.	1. Columns are well identified.			3
		2. Height of footing A4 is accurately indicated.			2
		3. Width of footing A4 is accurately indicated.			1
		4. Length of footing A4 is accurately indicated.			1
		5. Height of footing A4 is accurately indicated.			2
	5.2. Section is well interpreted.	1. Level of the beam is identified.			2
		2. Level of the slab is identified.			2
		3. Level of the roof is identified.			2
	5.3. Site plan and	1. Setback of the building to the			2

	electrical drawings are well interpreted.	center of the road identified.			
		2. Number of sports rights is well identified.			3
Learning outcome 6: (5%)	6.1. Road layout drawing is well interpreted.	1. Road centreline is well identified.			1
		2. Elements of the road are well identified.			4
Learning outcome 7: (5%)	7.1. Drawing sheet is well presented.	3. Drawing frame is well drawn.			1
		4. Title block is well drawn.			1
		5. Title block is well filled.			3
Total marks		100			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

References:

1. B.S. 11922: Construction Drawing Practice
2. <https://www.bigrentz.com/blog/floor-plan-symbols>
3. Malik, R. S. (2004). "Civil Engineering Drawing": Draughtsman Civil Theory and Practical, (4th Ed.). New Delhi: New Asian Publishers
4. Venkata, E. (2009). Textbook of engineering drawing with AutoCAD, Fourth edition BSP BS: Sultan Bazar
5. ISO 2692:2006 Geometrical product specifications (GPS) — Geometrical tolerancing — Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR)
6. ISO 1101:2017 Geometrical Product Specifications (GPS) — Geometrical Tolerancing — Tolerances of Form, Orientation, Location and Run-Out